RangeVol: volatility from the day's high and low

The open, high, low and close of each day say more about volatility than the closes alone: six classical estimators that use them, and the log range of Alizadeh, Brandt and Diebold.

In Pollis	The *Measuring Volatility* panel, in Extend > Financial Econometrics, the first panel of the Financial Econometrics Toolbox.
Missing in Julia	the range-based estimators of Parkinson, Garman and Klass, Rogers and Satchell, and Yang and Zhang, with close-to-close as the baseline, for daily OHLC prices. R has them in TTR's `volatility()`; Julia had none.
Package	`RangeVol.jl`, from GitHub: `Pkg.add(url = "https://github.com/Trumpingtons/RangeVol.jl")`.
Tests	38 tests: 18 against TTR on three datasets (AAPL, the S&P 500, simulated), 15 on simulated prices, 5 of argument checks. Recomputed today, the six estimators agree with TTR 0.24.4 to 2.0×10^{-14} or closer.

Why a package

The usual estimate of volatility is the standard deviation of daily returns, close to close. It ignores what happened during the day: a day that ends where it started may have travelled far in between. Parkinson (1980) showed that the high and low of the day, in a random walk, estimate the variance about five times as efficiently as the close; Garman and Klass (1980) added the open and the close; Rogers and Satchell (1991) allowed for a drift; Yang and Zhang (2000) allowed for the move overnight, between one close and the next open. Daily OHLC prices are what most data sources give, so these estimators come at no cost in data. Pollis' panel needed them in Julia, with the same conventions as TTR, so that a result can be checked in R.

Where they are used

- **Historical volatility** of stocks, indices, currencies and commodities, for risk measures and option pricing, when intraday data are not available.
- **Short windows.** With five times the efficiency, a month of ranges estimates the variance about as well as five months of closes.
- **Volatility models.** The log range is a nearly Gaussian measure of volatility for fitting stochastic volatility models (Alizadeh, Brandt and Diebold 2002).

The method

For n days with open O_t, high H_t, low L_t and close C_t, each function returns $\sqrt{\hat{\sigma}^2 \cdot k}$, where $\hat{\sigma}^2$ is the daily variance and k is `trading_periods`: 1 by default, for daily volatility; 252 to annualise a stock's, as TTR's `N = 252` does.

Function	Daily variance
`close_to_close(c)`	$\frac{1}{n-2}\sum_{t=2}^{n}(r_t - \bar{r})^2$, with $r_t = \ln(C_t/C_{t-1})$
`parkinson(h, l)`	$\frac{1}{4n\ln 2}\sum_t \ln(H_t/L_t)^2$
`garman_klass(o, h, l, c)`	$\frac{1}{n}\sum_t\left[\frac{1}{2}\ln(H_t/L_t)^2 - (2\ln 2 - 1)\ln(C_t/O_t)^2\right]$
`rogers_satchell(o, h, l, c)`	$\frac{1}{n}\sum_t\left[\ln(H_t/C_t)\ln(H_t/O_t) + \ln(L_t/C_t)\ln(L_t/O_t)\right]$
`garman_klass_yang_zhang(o, h, l, c)`	$\frac{1}{n-1}\sum_{t=2}^{n}\left[\ln(O_t/C_{t-1})^2 + \frac{1}{2}\ln(H_t/L_t)^2 - (2\ln 2 - 1)\ln(C_t/O_t)^2\right]$
`yang_zhang(o, h, l, c)`	$\hat{\sigma}_o^2 + \kappa\,\hat{\sigma}_c^2 + (1-\kappa)\,\hat{\sigma}_{RS}^2$ over $t = 2, \ldots, n$, with $\kappa = 0.34/(1.34 + \frac{m+1}{m-1})$, $m = n - 1$; needs $n \geq 3$

In Yang and Zhang's estimator, $\hat{\sigma}_o^2$ is the sample variance of the overnight returns $\ln(O_t/C_{t-1})$, $\hat{\sigma}_c^2$ that of the open-to-close returns $\ln(C_t/O_t)$, and $\hat{\sigma}_{RS}^2$ the Rogers–Satchell variance; κ is the weight that minimizes the variance of the estimate. Parkinson's and Garman and Klass's estimators assume no drift and no move overnight; Rogers and Satchell's allows for a drift; the last two include the overnight move, and Yang and Zhang's allows for both.

The log range

`alizadeh_brandt_diebold(h, l)` returns a vector: the log range of each day, $\ln(\ln H_t - \ln L_t)$, the logarithm of the range of the log price (Alizadeh, Brandt and Diebold 2002). It is close to Gaussian, with a mean that moves one for one with $\ln \sigma_t$, which makes it an observation of volatility for a state-space model; a day whose high equals its low gives $-\infty$, to be

dropped. Over 100,000 simulated days of 390 steps, the log range had skewness 0.139 and kurtosis 2.758, with a mean of $0.384 + \ln\sigma$ and a standard deviation of 0.299; the range itself, `exp.(alizadeh_brandt_diebold(h, l))`, had skewness 0.962 and kurtosis 4.176.

A worked example

The S&P 500 in 2023

The `sp500` dataset of PollisDatasets.jl has the daily open, high, low and close of the S&P 500 since 1985 (Yahoo! Finance, CC0); 2023 has 250 trading days.

```
using RangeVol, PollisDatasets, Dates
d = dataset("sp500"); i = year.(d.date) .== 2023
o, h, l, c = d.open[i], d.high[i], d.low[i], d.close[i]
yang_zhang(o, h, l, c)                                  # 0.007551, daily
yang_zhang(o, h, l, c; trading_periods = 252)         # 0.119866, annualised
```

Annualised, $k = 252$	close	Parkinson	G–K	R–S	GK–YZ	Y–Z
S&P 500 2023, 250 days	0.130996	0.108409	0.105824	0.103867	0.120032	0.119866
AAPL 2023, 250 days	0.199154	0.178093	0.178875	0.179683	0.213309	0.213400

For the S&P 500, the three estimators that see only the trading day give 0.104 to 0.108; those that add the move overnight, 0.120. For AAPL, from Yahoo! Finance through R's quantmod, the first three give about 0.179 and the last two 0.213, above close-to-close's 0.199: a single stock moves more between one close and the next open, as earnings and news released outside trading hours would make it, and the first three miss it.

What the data must be

The estimators assume that the open, high, low and close come from one session. Before 2006, the open of the S&P 500 in Yahoo's data is usually a copy of the previous close, so the move overnight is zero and the open-to-close return is the whole day's: use the range-based estimators that need the open only from 2008 on. Some sources of currency prices set the close equal to the open; check before trusting a range-based estimate.

What the tests show

Against R

`test/generate_data.R` downloads AAPL from Yahoo! Finance and the S&P 500 of 2023 from PollisDatasets.jl, simulates a third series, and saves TTR 0.24.4's `volatility()` of each with `N = 252`, over the whole series (`n = nrow` for the first four, `n = nrow - 1` for the two with an overnight term, whose window counts pairs of days). The estimates RangeVol computes today differ from TTR's by at most 2.0×10^{-14}, relative, in each of the 18; running TTR again in R gives the saved values to about 15 digits. The test suite computes RangeVol's estimates from the saved prices and compares them with TTR's at a relative tolerance of 10^{-10}.

By simulation

`vignette/simulation.jl` draws prices whose log moves as a random walk, with a daily standard deviation of 0.0126, in 390 steps a day (one a minute) or 23,400 (one a second), with or without a move overnight; 2,000 samples of 250 days, or 400 at seconds. The table gives the mean of each daily variance over the true one, and the efficiency: the variance of close-to-close's estimate over the estimator's, each divided by its mean.

	close	Parkinson	G–K	R–S	GK–YZ	Y–Z
Minutes, no move overnight						
mean / true	1.003	0.936	0.910	0.910	0.910	0.923
efficiency	1	4.65	7.21	5.64	7.18	6.78
Minutes, a fifth of the variance overnight						
mean / true	1.001	0.749	0.728	0.728	0.928	0.938
efficiency	1	4.50	6.96	5.45	7.61	7.24
Seconds, no move overnight						
mean / true	1.001	0.994	0.991	0.991	0.991	0.992
efficiency	1	4.68	6.98	5.98	6.95	6.81

The efficiencies are close to those the papers derive for continuous prices, about 5.2 for Parkinson's, 7.4 for Garman and Klass's and 6 for Rogers and Satchell's. A high and low observed once a minute fall short of the true ones, and the range-based estimators then understate the variance by 6 to 9 per cent; once a second, by less than 1. With a move overnight, the first three miss it, as in AAPL, and only the last two stay close to the truth.

The rest

`test/runtests.jl`: 15 tests that each estimator is positive and within 10 per cent of the volatility of 500 simulated days, each drawn in 390 steps with its high and low taken from the path, and that the log range is the logarithm of the range; and 5 that wrong arguments (vectors of different lengths, too few days) throw.

Still open

- **Rolling windows.** TTR's `volatility()` returns an estimate for each window of n days; RangeVol returns one for the whole series.
- **Bias corrections.** The discrete-sampling bias above has known corrections, and Yang and Zhang's κ assumes that the drift is small; neither is offered.

References

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Software: Ulrich, J., TTR, and Ryan, J. A. and Ulrich, J. M., quantmod (R); PollisDatasets.jl; CSV.jl, Tables.jl, StableRNGs.jl and Statistics (MIT).

Package written and tested with Claude Code (Anthropic) in Pollis.

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